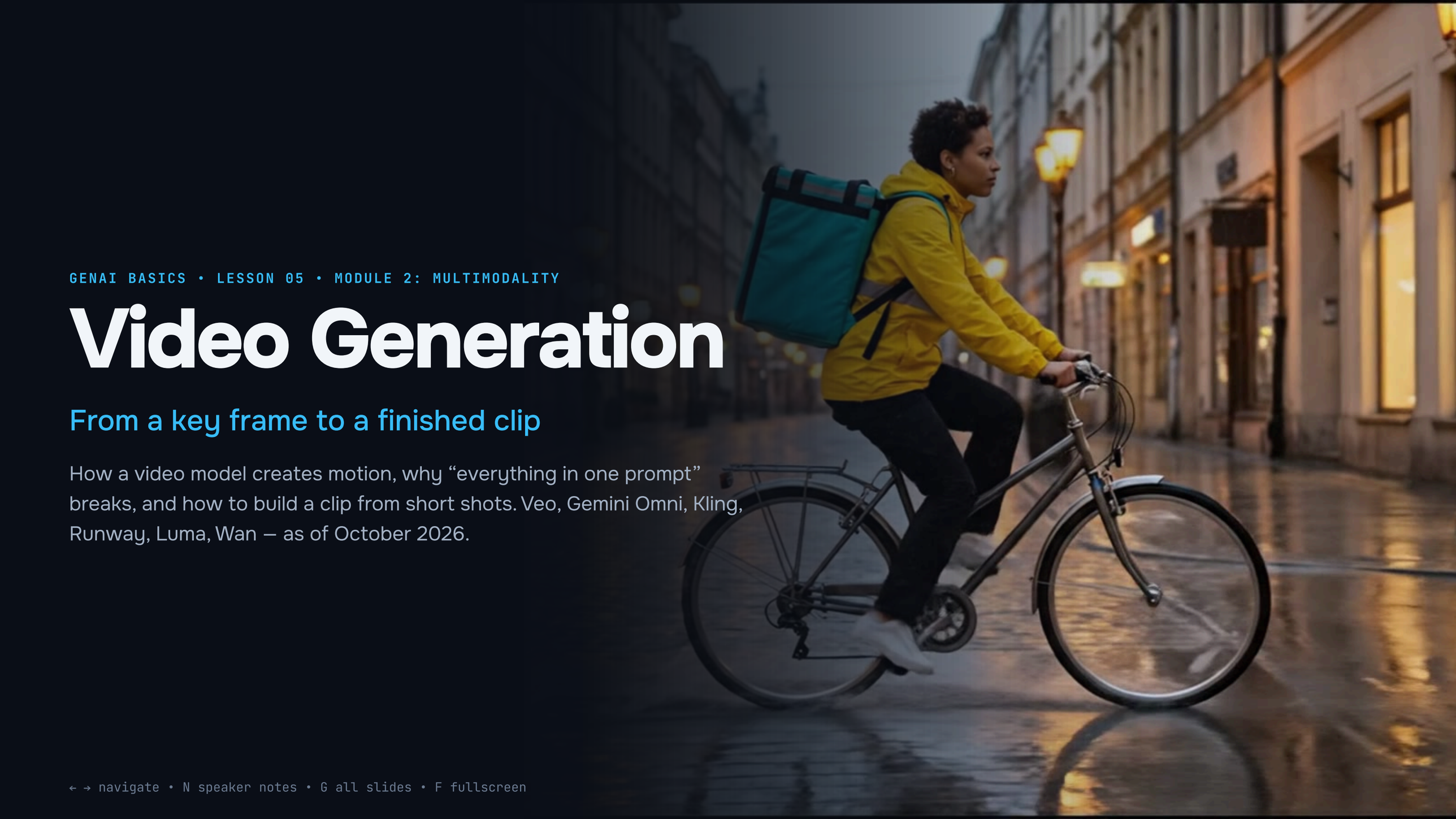
GENAI BASICS • LESSON 05 • MODULE 2: MULTIMODALITY

Video Generation

From a key frame to a finished clip

How a video model creates motion, why “everything in one prompt” breaks, and how to build a clip from short shots. Veo, Gemini Omni, Kling, Runway, Luma, Wan — as of October 2026.



START WITH THE RESULT

This clip was built from three 8-second clips



3 shots

each made from a **ready key frame** (image-to-video)

8+8+8 → 23 s

joined with two 0.5 s transitions

≈ $10

3 × 8 s × $0.40 — **Veo 3.1** list price, 720p, retries not counted

> Today we take it apart step by step. And we start with a failure: what happens if you ask for all of it in **one prompt**.

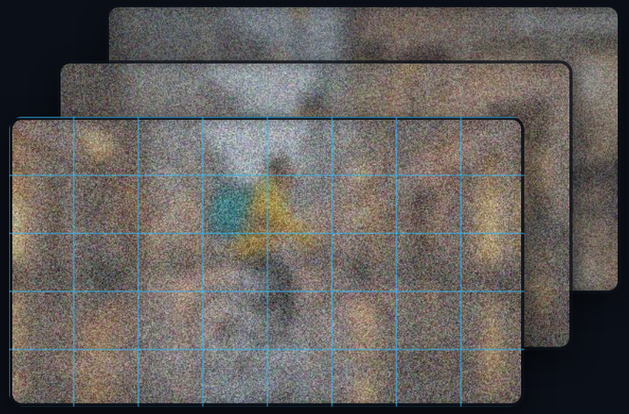
UNDER THE HOOD • NO FORMULAS

How a model makes a video



1. Noise

Start: random noise for **all frames** of the clip at once



2. Step-by-step cleanup

The video is cut into pieces “place in frame × moment in time”. Dozens of steps remove noise; the text and the first frame guide every step



3. Frames

24 frames per second that agree with each other — that is why motion looks smooth

There is no physics engine inside. The model has seen millions of videos and knows what motion *usually looks like*. So it fails where exact rules matter: **hands and objects in contact, text, long scenes**.

Schematic: frames from our shot 2, noise added for illustration.

CASE • FAILURE

Everything in one prompt: 4 typical failures

TEXT-TO-VIDEO Veo 3.1 • 8 s • text only



PROMPT (SHORTENED)

```
Our friendly Glovo delivery courier enters a cozy
Scandinavian cafe, … picks up a brown paper food delivery
bag, puts it securely inside her turquoise backpack, then
cuts to her riding a city bicycle through the rainy avenue
at dusk … and then camera zooms in to a smartphone map …
```

> One cause: the model got **3 scenes and 5 actions** for 8 seconds.



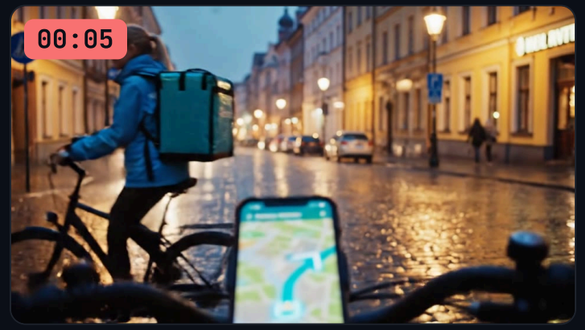
Action skipped

the bag stays on the counter — it never goes into the backpack



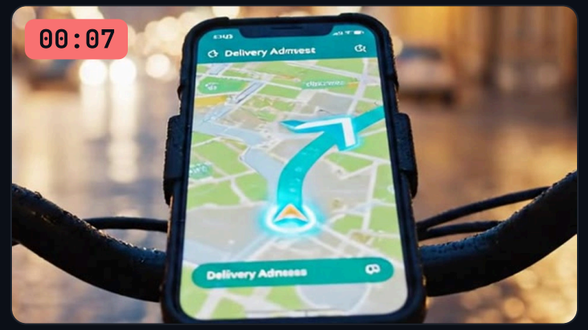
Details drift after the cut

the logo on the box changes: “Glono” → “Glovo”



Space breaks

handlebars with a phone slide in while the courier is still riding ahead



Pseudo-text

“Delivery Admest” instead of “Delivery Address”

THE KEY TECHNIQUE

First the frame, then the motion

T2V Text-to-video: text only



Text → The model decides **what is in frame** and **how it moves** → ✗ **two tasks at once**

I2V Image-to-video: ready frame + text



Frame ✓ + text about motion → The model decides only **how it moves** → ✓ **one task**

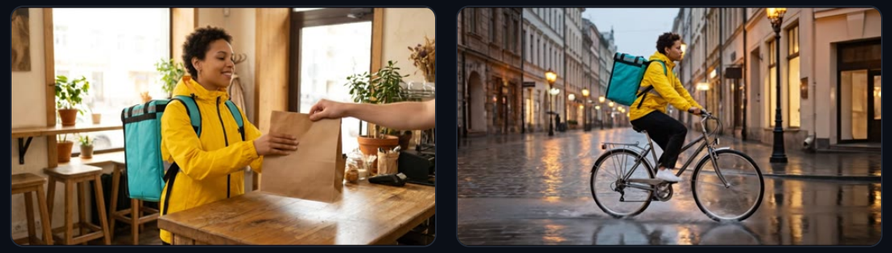
Rule: lock appearance, light and composition in an image (the L04 skill). Leave only motion to the video model.

STEP 1 • REFERENCES

A character sheet keeps the hero consistent



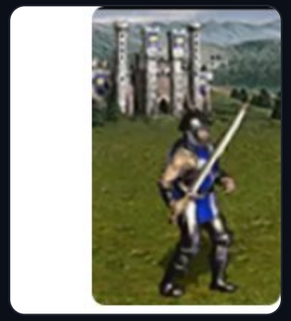
- A reference **narrows the model's choices**: less room to "invent" the look.
- **Veo 3.1** — up to 3 reference images; **Kling 3.0** — Elements, 2–4 images.
- More stable, but **not guaranteed** — check every shot by eye.



Result: **the same courier** in shots 1 and 2.

STEP 2 • KEY FRAME

Perfect the key frame in chat, not in video



Sprite — game source art



“Make a realistic 16:9 frame from this sprite”

Draft — wrong helmet and tabard



“Bare hands, tabard in white and blue rectangles, a flat ‘kettle’ helmet”

Approved ✓ — the first frame for video

seconds · cents

one **image** edit in chat

minutes · $0.40/s

one **video** regeneration (Veo 3.1, 720p)

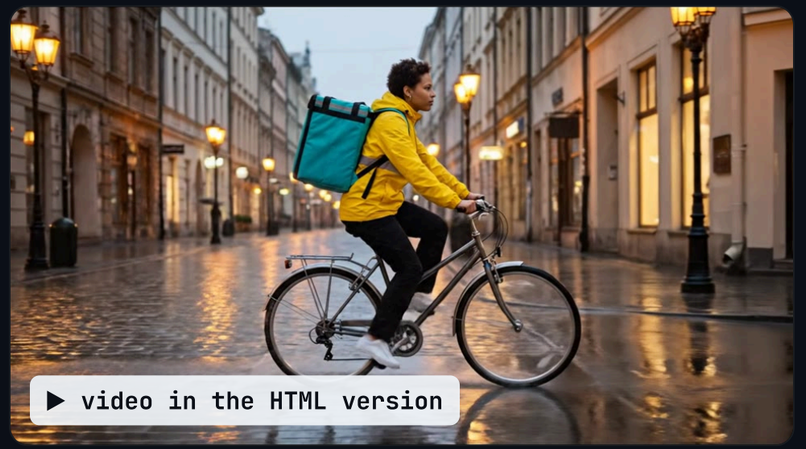
STEP 3 • PROMPT

The director’s prompt: 6 motion slots

1 Camera · 2 Who · 3 Action · 4 Where & when · 5 Motion details · 6 Light

ACTUAL PROMPT OF SHOT 2 (VEO 3.1, I2V)

1 Dynamic tracking shot moving forward alongside the courier. 2 The same courier with short cropped curly hair 3 pedals her bicycle forward 4 through the rain-slicked city avenue at twilight. 5 Wheels spinning smoothly forward with water spray cutting through shallow puddles. 6 Wet cobblestones reflecting warm shop lights and streetlamps.



Result — shot 2

- **Motion verbs:** pedals, spinning, tracks — the model needs to know what changes.
- **Don’t retell the frame:** it is already there. Appearance — one short anchor so it doesn’t drift.
- **Say what should happen.** A separate negative prompt field exists only in some tools (Kling, Wan).

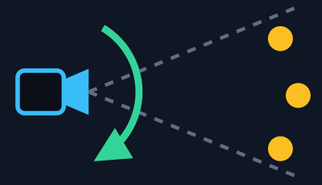
STEP 3 • CAMERA

8 camera moves that models understand



Locked-off camera `Static`

“static camera, locked-off shot”



Turn in place `Pan`

“slow pan left across the street”



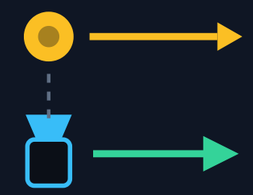
Tilt up / down `Tilt`

“tilt up to the rooftops”



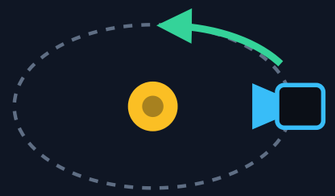
Move closer / away `Dolly in / out`

“slow dolly in to her face”



Move alongside the hero `Tracking`

“tracking shot alongside the courier”



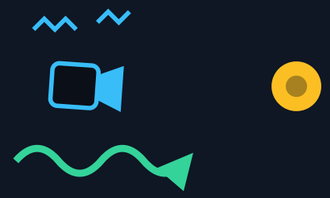
Circle around the subject `Orbit`

“slow orbit around the mug”



Camera rises up `Crane`

“crane up over the street”



Handheld, slight shake `Handheld`

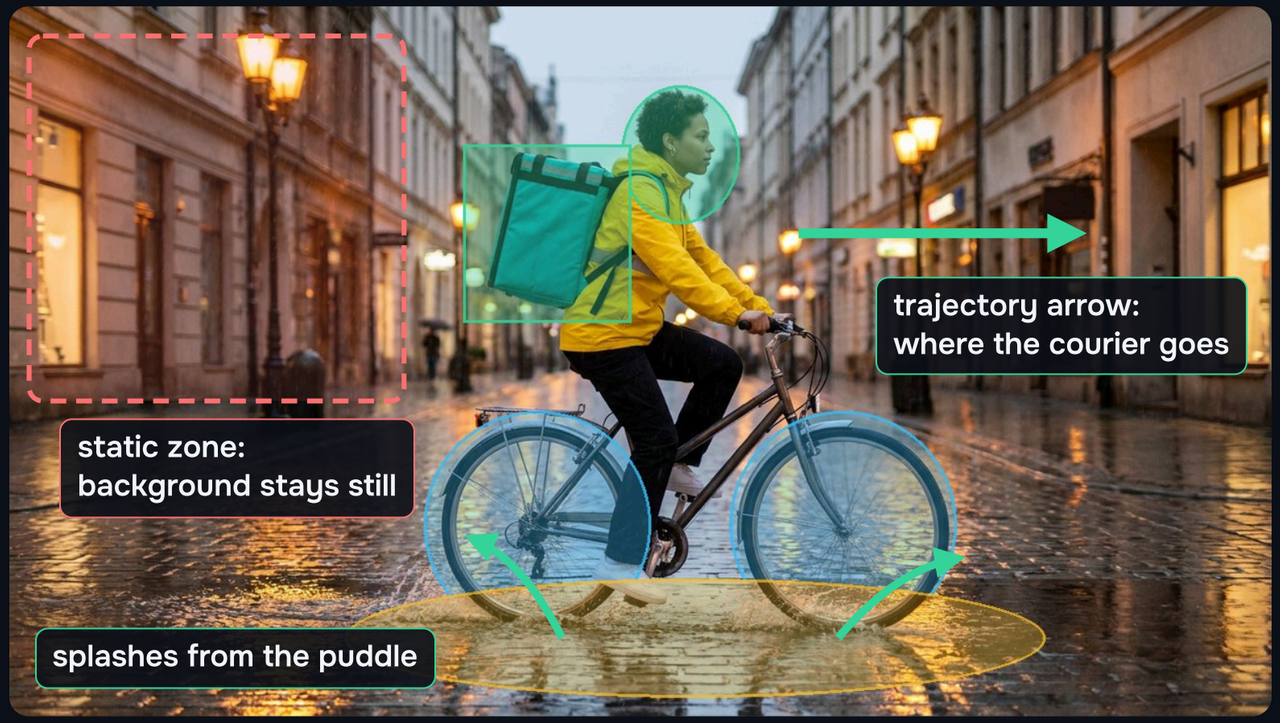
“handheld, slight shake”

> Models were trained on captioned video, so they understand **camera terms**. In chat apps you set the camera with words only; video studios also have dedicated controls (Runway, Kling, Luma).

Diagrams: top view (Pan, Dolly, Tracking, Orbit) and side view (Static, Tilt, Crane). Camera is blue, subject is yellow, motion is the green arrow.

STEP 4 • PRECISE CONTROL

What you can control more precisely than with words



Motion Brush (Kling): up to 6 selected elements, each with its own arrow, plus a static zone. Available on Kling 1.x–2.x models; not in the Kling 3.0 feature list.



First + last frame

The model fills in the motion between two images. **Veo 3.1, Gemini Omni, Kling 3.0**



Extend a clip

Veo 3.1: +7 s per step, up to 148 s (720p, not Lite)

Gemini Omni: +10 s per step, up to 40 s



Hero references

Veo 3.1 / Fast: up to 3 images, 8 s clip

Kling 3.0 Elements: 2–4 images

STEP 5 • EDITING

Short shots + editing instead of one long take



8 + 8 + 8 − 0.5 − 0.5 = 23 s

three Veo 3.1 clips and two transitions: a wipe and a fade through black

> You can join clips in any editor (CapCut, DaVinci Resolve) or with one `ffmpeg` command. A person does the editing, not the model.



BREAK

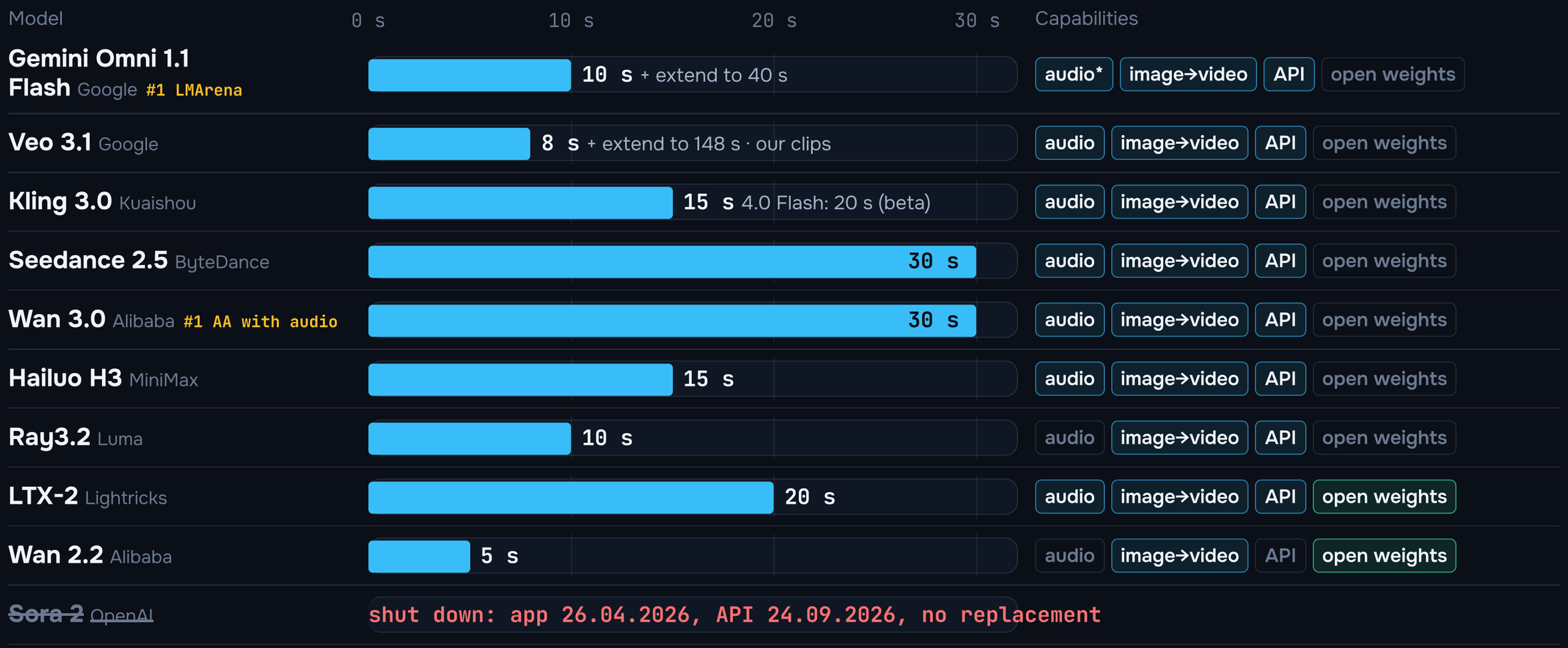
5 minutes for coffee and resting your eyes

05:00

Next: the model market, what it looks like in code, what it costs, how to check video and what the risks are.

MARKET • OCTOBER 2026

Video models: length of one clip and what they can do



Bar = maximum per generation. Highlighted chip = yes; grey = no or not confirmed. * Omni audio — per third-party reviews. Rankings: LMArena 21.09.2026, Artificial Analysis (with audio) — October 2026. Runway and Leonardo are platforms that host many of these models.

FOR REFERENCE • GOOGLE-GENAI

What it looks like in code

```python
import time
from google import genai
from google.genai import types

client = genai.Client()  # key from GEMINI_API_KEY
op = client.models.generate_videos(
    model="veo-3.1-fast-generate-preview",
    prompt="Tracking shot alongside the courier",
    image=types.Image.from_file(location="shot2.png"),
)
while not op.done:  # wait: ~11 s to ~6 min
    time.sleep(10)
    op = client.operations.get(op)
video = op.response.generated_videos[0].video
client.files.download(file=video)
video.save("shot2.mp4")
```

Three steps: **send → wait → download**. Video takes a while, so the answer does not come back immediately.

PRICE PER SECOND, 720P



Gemini API, October 2026. No free tier for Veo; you pay only for successful videos.

ECONOMICS

Count seconds, not attempts

A clip like ours: **3 shots of 8 seconds**, Veo 3.1, 720p.



The approved frame saves money: fewer tries per shot. Images for key frames cost cents.

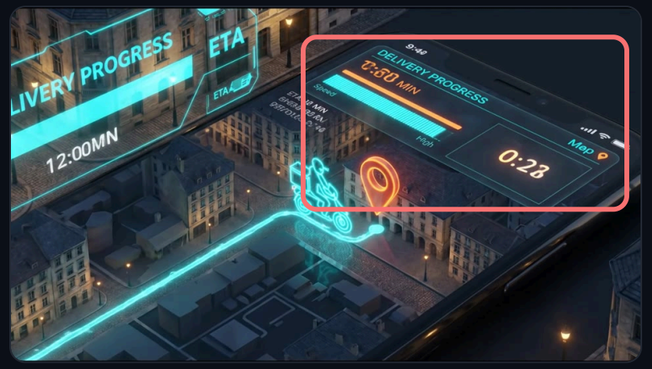
Drafts on a cheap model: Veo 3.1 Lite — $0.05/s, Gemini Omni 360p — about a third of the 720p price.

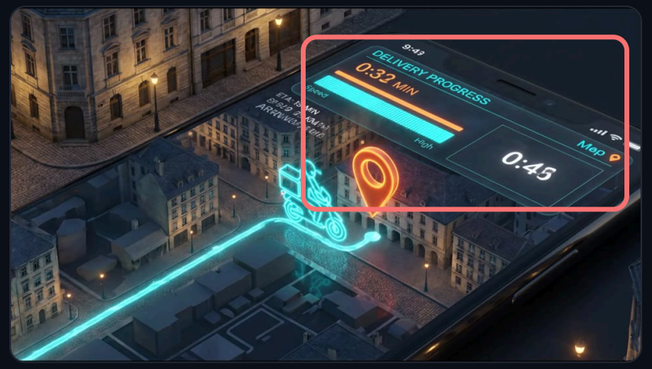
The number of tries is an assumption for the estimate. Gemini API prices as of October 2026; only successful generations are billed.

QUALITY CHECK

Checking a video is a separate step

Even a good I2V clip needs a frame-by-frame look. Defects from **our own** shots 2 and 3:





Shot 3, 1.5 seconds apart: the timer jumps 0:28 → 0:45, “Mep” instead of “Map”, the icon is a moped while the courier rides a bicycle.



Shot 2: the “THABIK” sign is random letters.

ACCEPTANCE CHECKLIST

1. **Hands and objects** in contact — don’t merge or vanish
2. **Text and numbers** — readable and stable
3. **Face and clothes** — match the reference
4. **Motion** — right speed and direction (wheels forward)
5. **First frame** — matches the approved one
6. **Shot joins** — light and colour match

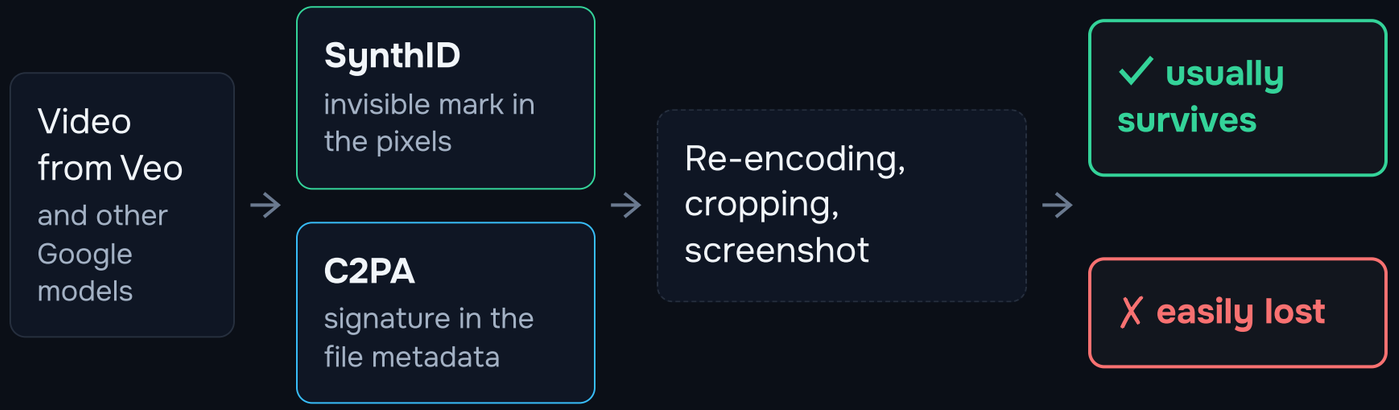
> This checklist is also the **homework rubric**.

RISKS AND PROVENANCE

Who made this video — and can you prove it

- **Your own frames and characters**
  OK — with an “AI-generated” label
- **Other brands, logos, styles**
  only with permission and after company lawyers review
- **Real people without consent**
  no: employees, clients, public figures — only with written consent

HOW A MARK SURVIVES RE-SAVING



SynthID check: upload the video to the Gemini app (up to 100 MB / 90 s). It only finds SynthID — videos from other services may go unrecognised.

EU AI Act, Art. 50: since 2 August 2026: **50(4)** — whoever publishes a deepfake must label it; **50(2)** — the model provider adds a machine-readable mark (systems released earlier: from 2 December 2026). Fines up to €15M or 3% of turnover.

PRACTICE • HOMEWORK #5

Bring an L04 frame to life



Track A · product

5-second promo: an orbit or dolly-in. The product's shape and text don't drift.



Track B · courier

A riding shot: the camera moves alongside. Face and uniform match the reference.

1. **T2V** in one prompt — for comparison
2. **I2V** with your L04 frame in **two services**
3. Prompts and settings → `video_spec.md`
4. Check against the slide 16 checklist → `video_qa.md`

```
genai-homeworks/L05/
├── videos/  # 01_t2v_trap, 02/03_i2v
├── video_spec.md
└── video_qa.md
```

Reviewer: **ihar_rubanovich@epam.com**. Free credits: Kling gives daily credits; check the other services before you start (see the homework file).

SUMMARY • BRIDGE TO L06

Four rules of video

1 **First the frame, then the motion**
I2V with an approved image instead of T2V

2 **One shot — one action**
4–8 seconds; the story is built in editing

3 **The director’s prompt**
camera + action + motion details

4 **Check and count**
a checklist for every clip; you pay per second



Lesson 06: sound and speech. Modern video models already generate sound themselves — how to control it is in L06.